Table 2. Controlled ablations with paired uncertainty. Removing components from the full system; each row is paired with the full system on the same eight seeds.

Configuration	Components				Paired evaluation		Latency
	Retrieve	Verify	Route	Cache	Success (%) mean ± SD	Δ vs. full (pp) [95% CI]	ms
Remove one component							
Full system	✓	✓	✓	✓	83.5 ±0.6	*NA*	512
Remove retrieval	✗	✓	✓	✓	75.2 ±0.7	−8.3 [−8.5, −8.0]	418
Remove verifier	✓	✗	✓	✓	78.8 ±0.5	−4.7 [−4.9, −4.5]	435
Remove routing	✓	✓	✗	✓	80.4 ±0.5	−3.1 [−3.3, −2.9]	566
Remove cache	✓	✓	✓	✗	83.5 ±0.6	0.0 [−0.1, +0.1]	689
Remove two components							
Remove retrieval + verifier	✗	✗	✓	✓	71.3 ±0.6	−12.2 [−12.4, −11.9]	341
Remove routing + cache	✓	✓	✗	✗	80.3 ±0.5	−3.2 [−3.4, −3.0]	748

Synthetic data. SD over 8 seeds; CI is a paired percentile bootstrap over seeds (20,000 resamples), so it excludes item-sampling uncertainty. pp = percentage points. Joint removals do not isolate single-component effects. Shaded: the reference for Δ. NA = not applicable. ✓ enabled, ✗ disabled.